

ReliefTrail - Transparent Disaster Relief Fund Tracking

Project Report

1. Introduction

Disaster response depends on timely support, but donors and communities can have limited visibility into how relief funds move. ReliefTrail is an EVM smart-contract prototype that records donations and payouts, and attaches a public cryptographic fingerprint to each payout. The fingerprint can be compared with an off-chain receipt or evidence file.

2. Problem Statement

Relief transactions may be recorded in private systems that outside observers cannot independently inspect. ReliefTrail demonstrates an append-only, shared transaction record with simple contract rules. It improves visibility into recorded payments but cannot verify a recipient's identity or prove that aid was delivered.

3. Objectives

- Accept donations and emit public donor/amount transaction events.
- Restrict payouts to the account that deployed the contract.
- Reject zero-value and over-balance payouts.
- Link each payout to an evidence file using a cryptographic fingerprint.
- Demonstrate deployment, contract state, events, and failed authorization in a local EVM.

4. Blockchain Platform and Tools

- Smart-contract language: Solidity 0.8.x (^0.8.24).
- Development environment: Remix IDE.
- Execution environment: Remix VM, a temporary local EVM with test accounts.
- Native currency: ETH in the example; amounts are represented in wei.

No public testnet or mainnet deployment is included. Remix VM account balances are for demonstration only.

5. System Architecture

```
Donor accounts -- donate(value) --> ReliefTrail contract -- DonationReceived event --> Public ledger
Organizer -- payRelief(recipient, amount, purpose, evidenceHash) --> Recipient account
                                     | checks organizer, amount, hash
                                     +-- ReliefPaid event + evidence fingerprint
Evidence document -- stored off-chain; its hash is recorded on-chain
```

The blockchain stores transactions and evidence fingerprints. The evidence file itself stays off-chain because putting receipts or personal data on a public chain would create privacy risks.

6. Smart Contract Design

Source file: contracts/ReliefTrail.sol.

Element	Purpose
reliefOrganizer	Immutable address of the deployer, allowed to pay out.
donate()	Receives a nonzero payment and updates donation totals and count.
payRelief(...)	Organizer-only payment; checks recipient, amount, balance, and nonzero evidence fingerprint.
availableBalance()	Returns funds currently held by the contract.
DonationReceived	Logs donor, amount, and resulting balance.
ReliefPaid	Logs recipient, amount, purpose, and evidence fingerprint.

The payout updates counters before transferring funds. If the transfer fails, the entire transaction reverts. The purpose and fingerprint are public. The contract is an early-stage prototype and has no independent evidence review or multi-signature governance.

7. Demonstration Procedure and Expected Outputs

In Remix, deploy to Remix VM. Donate 1 ETH from one test account and 0.5 ETH from another. From the deployer, pay 0.4 ETH (400000000000000000 wei) to a third test account. Include a nonzero 32-byte demo fingerprint in `payRelief`.

Check after sample transactions	Expected output
<code>totalDonated()</code>	1500000000000000000 wei (1.5 ETH)
<code>donationCount()</code>	2
<code>totalPaidOut()</code>	400000000000000000 wei (0.4 ETH)
<code>payoutCount()</code>	1
<code>availableBalance()</code>	1100000000000000000 wei (1.1 ETH)
Payout from non-organizer	Reverts with <code>OrganizerOnly</code>
Payout above available funds	Reverts with <code>InvalidPayoutAmount</code>
Payout with zero evidence hash	Reverts with <code>MissingEvidenceHash</code>

Live demonstration completed in Remix VM (Osaka). Deployed contract address:

0xd9145CCE52D386f254917e481eB44e9943F39138. The displayed state matched these values: 1.5 ETH donated across two donor accounts, 0.4 ETH paid out, 1.1 ETH remaining, donation count 2, and payout count 1. A non-organizer payout reverted with `OrganizerOnly`. The local Remix VM is temporary; this address is not a public network deployment. Capture screenshots from your own Remix session before sharing or submitting.

8. Conclusion

ReliefTrail demonstrates donations, an organizer rule, balance checks, transaction events, and evidence fingerprinting. It makes recorded fund movements easier to inspect. A matching hash can link a receipt file to a payment record, but it cannot establish that the receipt is genuine or that aid reached people.

9. Future Scope: Toward a Real Pilot

A future version could require multiple independent approvals, add recipient delivery acknowledgements, and provide a QR-based verification page. Receipts should remain encrypted off-chain, with only their fingerprints on-chain. Any real pilot should begin with a community or NGO partner, fictional/test data, independent security review, and appropriate legal and privacy safeguards.

References

- Solidity documentation: <https://docs.soliditylang.org/>
- Remix IDE: <https://remix.ethereum.org/>
- Ethereum developer documentation: <https://ethereum.org/developers/docs/>